

Learning Computer Architecture in Virtual Reality

* Required

* This form will record your name, please fill your name.

Screening and Background

1. Are you 18 or older? *

☐ Yes

☐ No

2. Have you previously studied computer architecture or computer organisation? *

☐ Yes

☐ No

3. Have you used a VR headset before? *

☐ Yes

☐ No

4. How experienced are you with Virtual Reality? *

☐ None

☐ Limited

☐ Moderate

☐ Good

☐ High

5. Before this session, how confident are you in your understanding of computer architecture concepts? *

- ☐ Extremely confident
- ☐ Somewhat confident
- ☐ Neutral
- ☐ Somewhat not confident
- ☐ Extremely not confident

6. How difficult do you usually find topics such as instruction flow, pipelining, stack behaviour, and memory hierarchy? *

- ☐ Extremely difficult
- ☐ Somewhat difficult
- ☐ Neutral
- ☐ Somewhat easy
- ☐ Extremely easy

7. Which of the following topics have you previously studied? *

Select all that apply

- ☐ Instruction Flow
- ☐ Pipelining
- ☐ Stack
- ☐ Memory Hierarchy
- ☐ Assembly
- ☐ None of the above

System Experience

Please answer the following questions about your experience.

8. Please indicate how strongly you agree with the following statements for System Usability *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The VR system was easy to learn	☐	☐	☐	☐	☐
I could navigate the environment without difficulty	☐	☐	☐	☐	☐
The controls felt intuitive	☐	☐	☐	☐	☐
The visual presentation made the concepts easier to follow	☐	☐	☐	☐	☐
I could focus on the learning task without being distracted by the interface	☐	☐	☐	☐	☐
I would feel confident using this system again	☐	☐	☐	☐	☐
I experienced discomfort, motion sickness, or visual fatigue while using the system	☐	☐	☐	☐	☐

9. Please indicate how strongly you agree with the following statements for System Engagement *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The experience held my attention	☐	☐	☐	☐	☐
I felt actively involved in the learning process	☐	☐	☐	☐	☐
The system made the topic feel more interesting	☐	☐	☐	☐	☐
I was motivated to explore the concepts further	☐	☐	☐	☐	☐
The interactive format improved my focus compared with standard learning materials	☐	☐	☐	☐	☐

10. Please indicate how strongly you agree with the following statements about your learning experience *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The VR experience helped me understand the topic better	☐	☐	☐	☐	☐
The visualisations helped me build a mental model of what was happening	☐	☐	☐	☐	☐
The system made abstract processes easier to understand	☐	☐	☐	☐	☐
I would prefer this as a supplement to traditional teaching materials	☐	☐	☐	☐	☐
I believe I learned something useful from this session	☐	☐	☐	☐	☐

Knowledge Check

Answer the following questions to assess your understanding of the computer architecture concepts covered in the session.

Reference Guide

- MIPS : Microprocessor without Interlocked Pipeline Stages, a Reduced Instruction Set Computer (RISC) architecture commonly used for teaching computer architecture.
- add: Adds the values from two registers and stores the result in a destination register.
- addi – Adds an immediate (constant) value to a register and stores the result.
- lw (Load Word): Loads a 32-bit word from memory into a register.
- sw (Store Word): Stores a 32-bit word from a register into memory.
- beq (Branch if Equal): Branches to a target address if two register values are equal.
- j (Jump): Unconditionally jumps to a specified instruction address.

11. Which register stores the address of the next instruction? *

- ☐ Stack Pointer
- ☐ Program Counter
- ☐ Instruction Register
- ☐ Base Register

12. Which stage of the instruction cycle reads an instruction from memory? *

- ☐ Decode
- ☐ Fetch
- ☐ Execute
- ☐ Write-back

13. During instruction execution, the instruction register primarily stores: *

- ☐ Program variables
- ☐ The currently executing instruction
- ☐ The stack pointer value
- ☐ Cache addresses

14. What happens to the program counter after an instruction is fetched? *

- ☐ It resets to zero
- ☐ It increments to the next instruction address
- ☐ It copies into the stack pointer
- ☐ It stores the result of the ALU

15. Which pipeline stage usually performs arithmetic operations? *

- ☐ Fetch
- ☐ Decode
- ☐ Execute
- ☐ Write-back

16. Branch instructions primarily affect which component? *

- ☐ Cache memory
- ☐ Program counter
- ☐ Stack pointer
- ☐ Instruction register

17. What does this instruction do? *add \$t0, \$t1, \$t2* *

- ☐ Adds \$t1 and \$t2 and stores the result in \$t0
- ☐ Adds \$t0 and \$t1 and stores the result in \$t2
- ☐ Moves \$t1 into \$t2
- ☐ Multiplies registers

18. What does this instruction do? *addi \$t0, \$zero, 10* *

- ☐ Stores 10 into \$t0
- ☐ Clears \$t0
- ☐ Moves \$t0 into \$zero
- ☐ Adds two registers

19. Which instruction type loads data from memory into a register? *

- ☐ ADD
- ☐ STORE
- ☐ LOAD
- ☐ JUMP

20. What is the role of register \$zero in MIPS? *

- ☐ Stores the program counter
- ☐ Always contains the value 0
- ☐ Stores return addresses
- ☐ Tracks the stack pointer

21. What does this instruction do? *lw \$t0, 0(\$s0)* *

- ☐ Stores \$t0 into memory
- ☐ Loads a word from memory into \$t0
- ☐ Moves \$s0 into \$t0
- ☐ Clears memory

22. What does this instruction do? *sw \$t0, 4(\$s0)* *

- ☐ Value from \$t0 is stored in memory
- ☐ Value from memory is loaded into \$t0
- ☐ \$s0 is overwritten
- ☐ Stack pointer moves

23. What does this instruction do? *beq \$t0, \$t1, label* *

- ☐ Jump if \$t0 > \$t1
- ☐ Jump if \$t0 == \$t1
- ☐ Jump if \$t0 < \$t1
- ☐ Always jump

24. What does this instruction do? *j label* *

- ☐ Jumps to label unconditionally
- ☐ Jumps only if two registers are equal
- ☐ Stores the return address in \$ra
- ☐ Loads a value from memory

25. Put the instruction stages in the correct order. *

Execute

Fetch

Write-back

Decode

26. How confident are you in your answers to the previous section?

1

2

3

4

5

Not at all confident

Very Confident

Open-ended Feedback

27. What part of the experience helped your understanding most? *

28. What part was confusing or ineffective? *

29. What would you improve in the system? *

30. Were there any moments where the VR interface got in the way of learning? *

31. Would you use this again as a learning aid? Why or why not? *

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